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The use of information technology for international transfer pricing in multinational enterprises

Lars Hemling [a], Jacob Christian Plesner Rossing [b,*], Andreas Hoffjan [a]

[a] TU Dortmund University, Faculty of Business and Economics, Germany
[b] The University of Tampa, Sykes College of Business, Department of Accounting, United States

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ABSTRACT

This paper studies the degree to which multinational enterprises (MNEs) use information technology for managing international transfer pricing (ITP). Based on 21 interviews conducted with in-house accounting and tax professionals in MNEs, we observed limited use of information technology for ITP management. However, some degree of ITP automation was observed in workflow management to produce transfer pricing documentation. The limited degree of automation observed was driven by both system- and individual-level barriers. Overall, we found that management accountants and information technology experts dominate the enterprise resource planning system design agenda, and the tax departments' ITP tax compliance objective plays a relatively limited role. This reduces the ability for ITP automation partly because the data segmentation that is prioritized for management reporting does not support the tax departments' needs for legal-entity data segmentation to document tax compliance.

1. Introduction

The tax strategies deployed by multinational enterprises (MNEs) continue to receive public attention. Several high-profile legal disputes between MNEs and tax authorities have demonstrated the significance of the tax practices of MNEs, particularly those related to international transfer pricing (ITP). ITP entails opportunities for tax avoidance resulting from the ability to exploit tax rate differentials between countries hosting affiliated companies. For example, an MNE with a company located in a tax haven can decrease its group-level tax burden by transferring goods at artificially high prices to an affiliated company located in a high-tax jurisdiction. This increases the selling company's profit, which is subject to limited taxation, while reducing the pre-tax profit of the buying company due to the inflated cost. The net effect is a reduced tax payment for the MNE group.

In response to alleged tax avoidance by MNEs, prominent trade institutions, such as the European Union and Organisation for Economic Co-operation and Development (OECD), exert continuous efforts to enhance global tax policy and promote tax compliance. In particular, the OECD and its recent action plan on base erosion and profit shifting (BEPS) have caught the attention of MNEs' top managers and tax departments. The BEPS action plan was initiated in 2013 by the OECD based on the G20 leaders' push for a supranational initiative to mitigate tax avoidance. The final BEPS reports (OECD, 2015) address various tax issues, including revised guidelines for setting and documenting transfer prices based on the arm's length principle.

Given the tightened tax regulatory environment facing MNEs combined with the increased attention to tax digitization in the tax

* Corresponding author.
E-mail addresses: lars.hemling@tu-dortmund.de (L. Hemling), jrossing@ut.edu (J.C. Plesner Rossing), andreas.hoffjan@tu-dortmund.de (A. Hoffjan).

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departments of MNEs (EY, 2017), the issue of information technology and automation relating to ITP are becoming increasingly important. More fundamentally, ITP is inherently part of MNEs' information technology environment, as intercompany pricing—similar to third party transactions—is typically processed within the enterprise resource planning (ERP) system. Academic studies and professional surveys of ITP have demonstrated how MNEs have responded to increasing tax regulatory pressures by deploying ITP strategies focused on tax compliance (Cools et al., 2008; Cools and Plesner Rossing, 2021; EY, 2019; Hummel et al., 2019; Klassen et al., 2017). However, the degree to which MNEs utilize information technology to manage ITP tax risks, particularly the extent to which MNEs depend on technology for pricing and documenting[1] intercompany transactions to meet tax regulatory requirements, has received surprisingly limited attention. For example, the degree to which MNE tax departments rely on manual processes and Excel spreadsheet solutions to manage ITP is unclear. At the same time, accounting and tax service providers are marketing various ITP software applications, which suggests some MNE interest in these applications. Overall, the degree of and motives for the adoption of these tools by MNEs remains unexplored in the academic literature, despite ITP tax risks receiving unprecedented organizational attention (EY, 2016, 2019; Klassen et al., 2017). Therefore, we seek to answer the following research question: *How are MNEs using information technology to manage ITP?*

Our paper contributes to a stream of contingency theory-based literature in accounting in which researchers interact directly with the MNE tax department professionals to understand how they adapt to the current tax environment (Cools et al., 2008; Cools and Slagmulder, 2009; Hasseldine et al., 2011; Hummel et al., 2019; Mulligan and Oats, 2009, 2016; Plesner Rossing, 2013; Rogers and Oats, 2019). Based on 21 interviews conducted between August 2018 and January 2019 with in-house accounting and tax professionals employed by MNEs, we found only limited use of information technology for ITP management. The main automation that was observed relates to the management of intra-organizational workflows for developing local documentation in accordance with the comprehensive 2017 update to the OECD Guidelines.[2] The limited use of information technology for ITP is significantly driven by system-level conditions, including fragmented ERP system environments. However, functional hierarchies also play a role in the limited use of ITP automation that was observed. Specifically, ERP system design is based on the prioritization of management reporting needs, typically divisional or business unit-level reporting, by the management accounting and information technology functions within the MNE. Conversely, ITP tax compliance is based on legal entity-level reporting. These differing data segmentation needs ultimately reduce the tax departments' ability to automate ITP processes.

The remainder of this paper is organized as follows: Section 2 presents the literature review and theoretical guidance, Section 3 presents the research strategy and design, Section 4 presents the empirical findings emerging from analysis of the empirical data, and Section 5 includes a discussion of our observations and suggestions for future research.

2. Literature review and theoretical guidance

2.1. Transfer pricing literature

Transfer pricing has a direct impact on management control systems in both domestic and multinational organizations, as it impacts the entire accounting cycle from planning and budgeting to performance measurement (Merchant and Van der Stede, 2017; Zimmerman, 2020). Initial research focused on transfer pricing for the purpose of optimizing resource allocation (Hirshleifer, 1956) and ensuring integration between diversified units (Colbert and Spicer, 1995; Van der Meer-Kooistra, 1994; Watson and Baumler, 1975). As globalization increased, the tax dimension of transfer pricing started to receive attention from tax policy makers looking to mitigate tax avoidance. At the same time, academics observed tax avoidance through transfer pricing (Clausing, 2001, 2003; Harris 1993) based on quantitative data obtained from public financial databases, see Hanlon and Heitzman (2010) for a review, and Becker et al. (2020), Blouin et al. (2012), De Simone et al. (2017), and Klassen and Laplante (2012) for more recent contributions.

Another stream of literature using contingency theory identified structural variables for the design of MNEs' transfer pricing systems (Borkowski, 1990; Borkowski, 1992a; Borkowski, 1992b; Borkowski, 1996; Emmanuel and Mehafdi, 1994). Using survey instruments as the typical data collection method, the literature identified relationships between organizational structures and the choice of transfer pricing methods (cost, market, or negotiated prices) and the managerial priorities and strategic objectives of transfer pricing decisions (Cravens and Shearon, 1996; Cravens, 1997). A more recent stream of qualitative research (Cools et al., 2008; Cools and Slagmulder, 2009; Plesner Rossing, 2013) expanded on the initial findings within the contingency school of ITP research, demonstrating the value of case studies for understanding complex ITP dynamics in MNEs. Cools et al. (2008) found that a tax compliance strategy has a direct impact on the control system design, including planning, budgeting, and performance measurement. Similarly, Cools and Slagmulder (2009) showed that ITP tax compliance policies require a redesign of responsibility accounting structures, including a conversion of cost centers into pseudo-profit centers. Plesner Rossing (2013) illustrated how interactive controls are used to refine ITP strategies in a dynamic and highly uncertain tax environment, including the role of interorganizational network ties.

Recent research has identified a shift in the strategic tax objectives of MNEs, and in many cases, previous tax avoidance practices

[1] Operational ITP involves two main tasks: a) setting transfer prices for intercompany, cross-border transactions based on the arm's length principle in the OECD Transfer Pricing Guidelines and b) producing documentation for tax authorities that demonstrates compliance with the arm's length principle.

[2] The revised OECD Guidelines (OECD, 2017) Chapter V requires MNEs to produce extensive transfer pricing documentation for business entities involved in intercompany transactions in order to demonstrate compliance with the arm's length principle.

have been replaced with more compliance-oriented strategies (Cools et al., 2008; Hummel et al., 2019; Klassen et al., 2017; Mulligan and Oats, 2009, 2016; Plesner Rossing, 2013; Wunder, 2009). These tax compliance strategies are typically driven by technical tax risks from double taxation as well as reputational risks (Austin and Wilson, 2017; Cools and Plesner Rossing, 2021). Moreover, research (Wunder, 2009) and professional surveys (EY, 2016, 2019) have identified tax risk management as a top priority for MNEs in transfer pricing. However, the use of technology for ITP in MNE tax departments, as well as tax departments' interaction with other key functions of accounting and information technology, needs to be better understood. In this context, the role of management accountants, including management accountants' transition from the traditional controller function to the role of strategic business partner for operational management, has been debated in accounting literature for decades (Byrne and Pierce, 2007; Horton and Wanderley, 2018; Shields, 1997). During this transition, it has become evident that management accountants play a key role in collaborating with in-house information technology staff to ensure that ERP system designs provide a basis for achieving the commercial goals of the organization (Grabski et al., 2011; Granlund, 2011; Rikhardsson and Yigitbasioglu, 2018). We believe the increasing importance of tax risk management in MNEs implies the need for a more direct interaction between the management accountants, information technology experts, and tax department professionals, which warrants further investigation.

2.2. Automation of accounting

Accounting and information technology are considered in a variety of literature streams, with most research residing in the accounting information system (AIS) and ERP system domains. The literature often applies a functionalistic approach, examining *system-level* drivers of information technology and ERP system adoption, design, and functionality. Some literature has advanced the understanding of the different pressures emerging from within and outside of organizations, driving the adoption of AIS systems and the degree of sophistication of these systems. For example, Kung et al. (2015) examined the influence of environmental factors on organization's intention to adopt specific software, using a taxonomy of mimetic, coercive, and normative pressures (Powell and DiMaggio, 1991). In context of technological adoption, mimetic pressures stem from isomorphic dynamics that drive imitation of peer organizations' technology systems and environments. Coercive pressures refer to formal or informal pressures for technological adoption from organizations upon which the adopter is dependent. Normative pressures involve forces driven by professional standards or professional communities within which an organization exists.

Other literature, including work emphasizing the importance of recognizing the *individual-level* drivers in explaining organizational practices and human behavior, has studied the impact of AIS and ERP systems on organizational decision making and control (Kocsis, 2019). Henri (2006) noted the importance of cultural variation across organizations to enhance the quality of performance management system design and managerial decision making. Other researchers have focused on the role of management accountants in the process of information system change (Byrne and Pierce, 2007; Granlund and Malmi, 2002; Jack and Kholeif, 2008; Malmi and Granlund, 2009; Quattrone and Hopper, 2005). Chapman (2005) noted that the theoretical potential of ERP system research goes beyond "technological curiosities" (p. 685) to explore and answer more fundamental issues of management control. Similarly, earlier work by Bariff and Galbraith (1978) suggested that the political significance of organizational reporting systems should be explicitly recognized in information system research, including the powerful coalition between organizational accountants and information system experts in the administration and distribution of organizational information through AIS and ERP applications.

However, despite an active AIS research agenda recognizing that all aspects of accounting data collection are impacted by technology (Arnold, 2018) and in some cases, controlled by technology (Sutton et al., 2016), research on the automation of accounting remains surprisingly limited. Kocsis (2019) reviewed current research and noted that while accounting professionals are seeing automation and artificial intelligence take over auditing and control, the academic literature does not reflect this empirical reality: "Automation only appears a few times in the [reviewed] articles, as does artificial intelligence" (Kocsis 2019, p. 8). Finally, in the context of transfer pricing, Taipaleenmäki and Ikäheimo (2013) conceptualized the recent convergence of management accounting and financial accounting with advancements in information technology, noting, "Within the technical and technological domain, recent developments in activities, such as financial reporting and stock market regulation, voluntary disclosures, performance measurement, transfer pricing[3], competitor, customer and contractor analysis, as well as due diligence in M&A activities, all contain major converging elements in which IT plays an important, even pivotal role." (p. 343).

Notwithstanding academics' and tax professionals' recognition of the inherent role of technology in tax risk management, few contributions have focused on the link between information technology and tax (Gallemore and Labro, 2015; Hamilton and Stekelberg, 2017). To our knowledge, no existing research, including research that interacts directly with the in-house accounting and tax professionals responsible for implementing MNEs' tax strategies, explores the degree to which MNE tax departments use technology to manage tax risks in general and ITP in particular. Therefore, in this paper we study MNE tax departments' use of information technology in the context of ITP and focus on the two main tasks of operational ITP, i.e., the implementation of the ITP policy through price setting and the subsequent development of documentation to demonstrate compliance with the arm's length principle. We examine two potential contingent factors of ITP automation: system-level drivers, e.g., ERP system design, and individual-level drivers, e.g., tax department managers' perceptions of ITP automation tools. Therefore, we use contingency theory as the theoretical basis for our analysis and the tax departments of MNEs as the level of our analysis. Fig. 1 illustrates the guiding framework for the empirical analysis.

[3] Emphasis added.

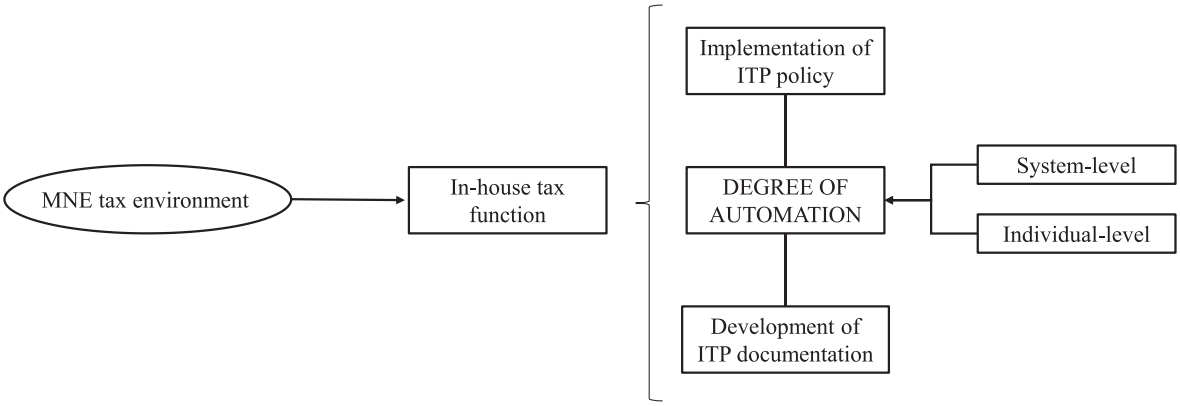

Fig. 1. Guiding framework.

3. Research strategy and design

This qualitative study focuses on 20 MNEs that operate in various industries and are all based in Germany, except one that is based in the US. The data were collected between August 2018 and January 2019 as part of a broader study on ITP and tax risk management in MNEs and include specific insights into ITP in the context of information technology as well as operational insights into the general management of ITP. The timing of the data collection provides a basis for making observations regarding the MNEs' perspectives and responses to OECD's BEPS initiative. The MNEs under study conduct high volumes of inter-company, cross-border transactions that interact with the ERP system by design. In addition, EY's transfer pricing survey found that Germany was the most frequently cited nation where transfer pricing policies have faced official examination (EY 2016, p. 18), creating a suitable context for our study.

Our analysis draws from a variety of data sources, and the main data input comes from 21 interviews with 25 in-house accounting and tax professionals.[4] These interviewees were carefully selected and either have direct ITP responsibility or at least functional involvement with ITP as part of their professional duties. This is in accordance with the "purposive" approach to sampling and interviewee identification recommended in the qualitative research methodology literature (Miles and Huberman, 1984, p. 27). Table 1 illustrates the characteristics of the MNEs and interviewees.

Prior to the interviews conducted by one of the researchers, the MNEs' websites were reviewed to ensure a sufficient degree of commercial understanding of the MNEs in question. In addition, publicly available documents from commercial vendors offering ITP software were reviewed, including the software's intended functionality for ITP automation. The total interview time amounted to 26 h and 59 min. Except for one interview in which detailed notes were taken, all interviews were recorded and transcribed. The interviews consisted of a pre-determined list of questions to guide the conversation (see Appendix A) but still allowed interviewees to elaborate on

Table 1
MNE and interviewee characteristics.

MNEs	MNE Size by Revenue (EUR Billions)			Σ	Σ
Sectors	<1	1–10	>10	MNEs	Interviews
Mechanical and Plant Engineering	1	8		9	9
Pharmaceutical Industry			3	3	3
Consumer Goods Industry			1	1	1
Retail and Wholesale		3	1	4	4
Industrial Conglomerate			1	1	2
Manufacturer Household Appliances		1		1	1
Manufacturer Electrical Material	1			1	1
Total	2	12	6	20	21

INTERVIEWEES	Headquarters	Subsidiary	Σ	Duration
Tax/TP manager	21	1	22	24h 42m
Internal auditor	1		1	0h 56m
Controller		2	2	1h 21m
Total	22	3	25	26h 59m

issues specific to their industry or unique to their organizational context (Yin, 2018). The interview guide was primarily based on the existing ITP literature, and also benefited from input from a Big Four partner, as well as insights obtained by one of the researchers who

[4] Two interviews had two interviewees participating, and one interview had three interviewees participating.

participated in a two-day network group meeting with in-house tax and accounting professionals.

One of the researchers coded the interviews based on a detailed coding scheme using a qualitative research analysis software tool (MAXQDA). The initial coding scheme was based on the themes and questions in the interview guide and the guiding framework in Fig. 1. The coding scheme was developed by two of the researchers through a highly interactive process based on careful review and analysis of the interview data. The coding of interviews followed a reconciliation approach to arrive at a mutually agreeable coding solution. Specifically, one of the researchers coded three interviews and shared the coding via MAXQDA with the other researcher who carefully reviewed the coding to check for consistency and completeness vis-à-vis the coding scheme. This process resulted in valuable conversations that guided the subsequent coding of the remaining interviews, and these coded interviews were also shared via MAXQDA for review before the joint analysis proceeded. The analysis of the coded interviews can be described as a highly interactive process to ensure alignment in the analytical process and enhance validity and reliability in the conclusions derived from iterative reflections. As part of the analytical process, a thematic matrix was developed based on the guidance of Miles, Huberman, and Saldana (2019). This matrix also served as the basis for the summary of the findings in Table 2.

4. Findings

4.1. The MNE tax function

The MNEs examined in this study were subject to increased attention from global tax authorities in recent years. In addition, the MNEs' boards and C-suite executives have started to interact more directly with their tax departments regarding tax risk management strategies, including those linked with ITP.

> *"This is high risk and therefore high value. They know BEPS[5]; they have heard from their network, forums, and their advisors. Therefore, they come to us and ask, 'How do we handle it?' That is actually also why you see BEPS everywhere in the compliance guideline because it is a game changer." (INT 9)*

Tax departments are typically organized as a subdepartment of the finance function, which handles group-level accounting, reporting, and compliance. Tax departments typically have international responsibility for direct and indirect taxes, including ITP, and generally apply a centralized approach to ITP tax risk management—a phenomenon that seems to be further accelerated by BEPS—to ensure consistent ITP control across the group. Only in a few cases, interviewees reported having dedicated tax people located at the subsidiary level. The main operational ITP responsibility of tax departments involves setting arm's length budget targets and corresponding transfer prices and producing the necessary documentation at year-end. More ad hoc ITP tasks handled by the centralized tax department relate to providing support to subsidiaries during local tax audits.

We observed that top management sets the tone that conveys the organizational appetite for tax risk to the tax department professionals responsible for the implementation of the tax strategy. This indicates a move away from tax being a decentralized compliance exercise receiving little active top management attention to an open-door policy relating to the tax strategy.

> *"Since BEPS, the door of the board of directors is always open to us. In the past, the topic (ITP) was still viewed very negatively, and we only did what was necessary to meet the legal requirements." (INT 6)*

As part of the transition towards implementing tax strategies for ITP in accordance with BEPS, we found that the vast majority of MNEs considered establishing some degree of automation for price setting and documentation. These ITP automation tools could be either a software solution that integrates with conventional ERP software systems or add-on software that extracts and processes data from the ERP system. Key software functionality for ITP price setting promoted by the external vendors includes the automated creation of segmented profit and loss statements and dashboard-based margin monitoring. Scenario-generating features that simulate the profit margin effects of specific ITP policies are mentioned in the marketing materials as being key features of ITP automation tools. For the automation of ITP documentation processes, the tools typically offer prebuilt text narratives to describe the business functions and risks of legal entities. In addition, several tools offer workflow process management to assist tax departments in obtaining feedback from key organizational functions responsible for providing inputs for completing local documentation files. This includes controllers, subsidiary finance managers, and business operational experts holding detailed transactional and operational knowledge to support ITP decisions. Section 4.2 and 4.3 consider the way in which the MNEs under study perceived these tools and the degree to which they are used for managing price setting and documentation for ITP.

4.2. ITP automation: Setting transfer prices

Regarding organizational structure, the German headquarters can generally be viewed as the main entrepreneur of the group that assumes significant business risks and owns the key intangible assets. However, to ensure operational and tax efficiency and due to historic acquisitions of companies holding significant intangibles, the group's intangible property (IP) is not always owned and maintained by the parent company.

[5] Base Erosion and Profit Shifting. For details on the OECD's BEPS initiative to reduce MNE tax avoidance, see the Introduction section. The specific BEPS reports that resulted from the OECD action plan can be accessed at https://www.oecd.org/tax/beps/.

"For example, if we had acquisitions and the acquired company had some kind of IP, which happened to be in a low-tax country, then the IP was left there, which then led to us having a lot of IP owners." (INT 11)

The main intercompany transactions observed were transfers of physical goods and services. In addition, the use of intangible property through intercompany licensing arrangements was emphasized.

"The most important cross-border supply and service relationships are…the tangible goods transactions, [such as] physical goods transactions in the various divisions of a company. Then there are IP-related transactions, such as a licensing system between affiliated companies. And the third major area is the clearing of services between affiliated companies, which is then split up into various sub-categories." (INT 25)

The standard approach observed for transfer price setting is one in which the tax department at headquarters develops an ITP policy for the group in cooperation with top management, primarily involving the CFO who typically assumes the ultimate tax responsibility. Interviewees stressed that the centralization of ITP is important to ensure a consistent approach across the group. Therefore, while managers of subsidiaries provide important operational information to the tax department in Germany, they generally have a limited influence over the formal ITP policy design. The policy typically highlights the tax compliance objective and ITP methods applied to specific intercompany transactions. It was evident from the interviewees that the tax department is deeply dependent upon other administrative functions to ensure the successful implementation of the policy. In particular, the accounting department and information technology specialists were emphasized due to their extensive involvement in handling policy implementation, including their assistance in ensuring that the methods selected by the tax department are applied to specific intercompany transactions.

"Systemic conversion takes place in SAP. This means that we pass information to colleagues in the shared service centers who then enter this information in the accounting organization (system) so that prices can be calculated correctly." (INT 25)

We found that one of the main issues in the ITP exercise that has significant implications for automating the process is ensuring useful data segmentation. Given the tax departments' need to document the arm's length nature of intercompany transactions based on ex-post accounting data, useful segmentation of intercompany transactions in the ERP system is a crucial prerequisite for automation.

"We actually want to define certain data that we need from the individual SAP ERP systems of the various companies…this is the only way to determine, for example, the segmented income statement." (INT 8)

We observed several limitations in the tax departments' ability to integrate ITP policies with existing ERP system designs. In general, interviewees agreed that the attention to ITP from top managers increased post-BEPS. However, the tax departments' ability to influence the ERP system design based on the data and segmentation needs for tax compliance remains rather limited. Instead, the ERP system design decisions are generally made by the management accountants and information technology specialists to ensure the ability to convert the ERP system information into a format that satisfies the reporting needs of the business and operational managers.

"We have to acknowledge that tax is not the center of the world. We follow business; business does not follow us." (INT 9)

In general, the accounting techniques applied in ERP systems for control and periodic management reporting are different to the segmentation needed for tax compliance. This includes accounting practices that are not necessarily aligned with the tax departments' objectives and rationale for what constitutes arm's length behaviors. For example, the management accountants and controllers perform cost allocations strictly for business decision making and reporting purposes without explicit regard to tax compliance, which can cause significant issues for the tax department, including the risk of double taxation.

"It is indeed the case that there is a field of tension between Controlling and the tax department, in the sense that Controlling is often interested in a very detailed, cause-related cost allocation…this applies particularly to services. In the case of material goods, there are normally not many problems, but in the case of intercompany service allocation, it is often the case that from the point of view of Controlling, an allocation is necessary, and the tax department or the transfer price team may then have to say, 'No, we can't do that easily.' or, 'It is not possible in this country. We can indeed offset, but then these are non-deductible business expenses.'" (INT 25)

A related issue that was found to limit interviewees' enthusiasm for automating the transfer price setting process is the fact that several MNEs use a variety of different ERP systems. One reason for these fragmented ERP environments is past mergers and acquisitions in which the acquired company's ERP platform remains in use after the acquisition. Such fragmented ERP structures create a barrier to ITP automation, as they require significant reconfiguration and standardization before an automated ITP solution can be implemented:

"In my view, the basic prerequisites for the transfer pricing process to be integrated into tax operations would be the standardization of ERP systems, which is not the case in all of our countries at present." (INT 17)

"A heterogeneous SAP landscape, that is where it starts." (INT 10)

However, this does not mean that MNEs' tax departments consider ITP automation solely an issue of ensuring more structure and uniformity in ERP systems. Some interviewees noted the importance of having a greater degree of standardization in the ITP policy throughout the MNE value chain. In fact, the standardization of the ITP policies and methods was considered a precondition for ITP automation to be possible.

"As soon as we can standardize, we can start automate." (INT 9)

"The simpler the transfer pricing system, the more it can be automated. If you have a transfer pricing system with a thousand exceptions, then in theory, you can still automate it, but it is difficult." (INT 11)

Interviewees generally seemed to be intrigued by the vision of ITP automation, with some being in the experimental phases of automating the pricing of specific intercompany transactions. This was perceived to be a long-term and costly project subject to

significant uncertainty because of system-based limitations.

"We are also talking about a period of probably ten years for implementation." (INT 16)

We observed only limited degrees of automation for the diagnostic control of profit margins relating to arm's length profit targets. One of the perceived benefits of these tools is the tax departments' ability to handle cases in real-time to which the management accountants were unsuccessful in implementing the ITP policy conveyed by the tax department. Some interviewees noted that while the ITP policy guiding intercompany invoicing was formalized at the beginning of the fiscal year, they would often experience significant variance from targets when reviewing the financial results.

"If you had detailed planning for the entities, then you could check during the year with margin-oriented transfer pricing methods whether I am heading in the right direction…then you could ideally make adjustment entries during the year and not only at the end of the year. This would mean less discussion during the tax audits. There are already these cases in which the budget values you offset against are not correct, and then there is, for example, another double offsetting at the end of the year. Then, of course, every auditor comes and asks, 'What is this? Why is this? So, is it really based on arm's length data or not?' That's always a topic of discussion." (INT 24)

For those relatively few tax departments that use monitoring tools to enhance their real-time arm's length-based diagnostic control, we found that the tools were not integrated with the ERP system but are add-on solutions that pulled data from the relevant ERP modules. The tax department subsequently processed the data to assess the degree of periodic adherence to arm's length earnings targets, as well as forecasting for the individual subsidiaries.

"With the help of this system, forecast values can also be determined for the profit margins, so to speak, which can then be applied for the following quarters on a certain observation date…It gets data from SAP. This data is imported separately, but we then report the calculation result back to our colleagues…who then enter it into the [SAP] system via the appropriate interfaces." (INT 25)

However, these solutions have not been integrated in a way that provides a basis for full ITP automation, including an automated journal entry adjustment within the relevant ERP system module. Instead, we found that manual assessment and data processing using Microsoft Excel is still widely practiced in tax departments. The outputs are then communicated to controllers who perform the necessary accounting journal entry adjustments in the ERP system for tax compliance purposes.

"The calculation process and simulation process is all done in Excel." (INT 19)

The general ITP automation vision of these MNEs is a solution in which the ERP system can handle the implementation of the ITP policy. However, for some MNEs it is not only a matter of automating the policy-based price setting process but also the ability to perform real-time simulations, including the ability to directly simulate pre- and post-tax effects from potential changes to transfer prices for specific transactions.[6] An interviewee currently involved in an ITP automation project commented:

"So, you can imagine, we have different SAP systems in use…and from these systems, it should be possible to import the individual tables from the financial, controlling, and materials management data into the uniform business warehouse in which it will be possible to monitor and adjust the margins in a permanent manner, and, of course, we also imagine that modulation calculations can be made on this basis. What will change if we modify a parameter for a specific country in a specific transfer pricing transaction? Then it can immediately calculate…what the effects will be. Then you can include the tax rate and so on. This is what we envision for the IT system at the end of the project." (INT 8)

Of note, we also received interesting feedback on the affinity for information technology among the tax department professionals. For example, one interviewee indicated a preference for playing a more limited role in relation to ERP system design to ensure that the scope of ITP responsibilities does not create an unrealistic burden for the tax department or distort operational control.

"We also have to think carefully about this from the [perspective of the] tax department: How much responsibility we can take; how much do we want to take on; how are our resources at that point. Because that is always my example or my position here that I represent in the company. We cannot build up a second controlling department with our resources in the tax department. That is simply not possible. There are only a handful of people who deal with transfer pricing in the company, but hundreds, if not thousands, of colleagues who deal with business analytics or controlling, and we cannot, so to speak, saddle up on that. We can't do that. You just have to work together, and the tax department has to think about what it is able to achieve with the human resources it has." (INT 25)

Along these lines, it seems that tax department professionals are mainly comfortable with more traditional domains of tax compliance, e.g., corporate tax law, accounting, and finance. One interviewee noted that tax department professionals are reluctant when it comes to becoming directly involved in more complex information technology and system applications.

"If they (tax departments) want to be perceived as doing the right thing, they have to work more with IT. This scares a lot of tax professionals because many of them are tax nerds. They do not like IT, and they do not understand IT. I actually look forward to the new generation of the so-called tax technologist; the guys who understand tax from a structured perspective and…are able to squeeze that into an IT system. This is what we will see. It [has been] stressed for the last couple of years, and it will be bigger and bigger. We need more IT systems and automation because our compliance burden will increase, and we will not get more people. That is the direction." (INT 9)

Furthermore, the importance of information technology in tax departments' recruitment decisions was highlighted. There appears to be an important generational factor at play here, as the younger staff who currently fill entry-level positions within tax departments

[6] For example, such analysis would be relevant under a transactional net margin method to test whether relevant earnings margins fall too far from a benchmark median value or a similar measure of arm's length comparability.

were reported to have more technological affinity.

> *"I have people in my team who not only [understand] tax law but also have knowledge of the systems, [*i.e.*], SAP and other IT-based systems. The whole department becomes much more technical. We are now reaching a technical level that the controlling department has developed for years. This is a good development…maybe it's also due to the low average age in the tax department that people there are more open to such topics." (INT 24)*
>
> *"My experience with the older colleagues here is that they have problems with Excel alone." (INT 13)*

Other interviewees echoed the importance of tax department professionals acquiring general information technology knowledge and a genuine interest in working with system applications to achieve ITP objectives.

> *"IT knowledge is definitely an advantage. There must also be an IT affinity, in other words, the willingness to work a lot with systems. In transfer pricing, you can't just rely on saying, 'I'm working on the topic from a legal perspective.'" (INT 25)*

4.3. ITP Automation: Documenting transfer prices

The MNEs under study are subject to increased documentation requirements, both in Germany and abroad, because of global tax authorities implementing the results of Action 13 in the BEPS action plan. The formal result of Action 13 was the introduction of a master/local file documentation approach accompanied by a country-by-country report to assist tax authorities in their risk assessment of individual MNEs. The master file causes only limited issues for MNEs, as it contains holistic, descriptive information regarding the value chain of the MNE and does not involve quantitative accounting data that is potentially subject to automation. However, the local documentation now requires more detailed information regarding specific intercompany transactions to justify accounting practices for ITP. MNEs generally manage the documentation process in-house and often use consultants to ensure that documentation files are aligned with local tax compliance requirements, although some MNEs with large tax functions utilize consultants less frequently.

> *"We do [the documentation] in-house. Consultants are minimally involved…because we have a very large tax department, with almost 40 people worldwide." (INT 1)*

Notably, the documentation exercise is not detached from the initial transfer price setting exercise (see Section 4.2). Rather, they were found to be closely linked, as the documentation essentially requires justifying the arm's length nature of transfer prices that have been applied to specific transactions. Given that transfer price setting is deeply integrated with the ERP systems, the tax department is forced to interact closely with the relevant ERP modules to extract and process accounting data for documentation purposes.

> *"In my opinion, transfer pricing is 80% IT because I must crystallize the transaction. I have to know how it was calculated [and] what margins are involved. In other words, very IT intensive in my opinion." (INT 6)*

One of the main obstacles observed to automating the ITP documentation process related to the lack of alignment between the segmentation required for management reporting versus for transfer pricing documentation purposes. Specifically, we observed that operational control and periodic management reporting were based on measuring commercial profitability at the divisional or business-unit level and were therefore relatively aggregated. In contrast, tax regulations require profitability to be documented at the transactional level for each legal entity. This difference in data segmentation causes a significant barrier to automating the production of documentation for the tax function, which was noted by several interviewees.

> *"The controllers often look at business units and not at legal entities. They also combine business units from different entities and…across borders. We (tax department) only ever look at if there is a company and if it settles with another company across the border and what the result is. This perspective is almost only needed in the tax area." (INT 24)*

That said, we do see a more widespread use of automation tools for ITP documentation than for price setting, and the interviewees exhibited a more positive attitude regarding the ways in which the tools from external vendors could assist the tax department in effectively completing documentation tasks. However, the lack of automation in the transfer pricing setting creates a barrier to the degree of automation in creating documentation owing to the aforementioned interdependencies. Therefore, the limited automation in price setting was found to limit the ability to automate the documentation process via direct data imports from the ERP system to the documentation files. Instead, a rather time-consuming and error-prone manual documentation process was observed, typically using Microsoft Excel for reorganizing ERP data imports. One of the popular features mentioned by the interviewees from MNEs that use documentation tools was the pre-narrated text modules that can be customized and reformatted to the specific commercial circumstances. Several MNEs were intrigued by these tools and had already implemented them or were actively planning to do so because they can enhance standardization in formatting the documentation reports.

> *"We have a layout. The [MNE name] local (documentation) file always looks the same, the structure is always the same so that we know afterwards where to find specific information." (INT 13)*

Several MNEs had previously developed fragmented and misaligned documentation in specific jurisdictions, making it difficult and time consuming to review and update the files. By using the template text modules from these tools, the local files can be created in a more consistent manner and revised as needed, typically by a local consultant who adjusts to the local documentation requirements. The increased documentation requirements that emerged from BEPS seem to be a key driver in switching from the manual, loose-leaf documentation approach to a more standardized and rigorous format by using documentation tools.

> *"We are currently in the process of introducing a new tool, which takes a much more centralized approach. Where we centrally issue transfer pricing templates from the tax department. Where we describe our main transactions in coordination with the business. We also*

do centralized competition and industry analyses, which we all feed centrally into this tool and actually tell the group companies that this is your basic framework in terms of content [and say], 'Please take a local consultant who will only adapt the local content to your needs.'" (INT 13)

"The best example is the documentation tool, which has simply not been an issue for us up until now. Before that we had a collection of sheets of paper that have been accepted by the German tax authorities in this format. We can no longer continue to use this format after BEPS, and either we do everything manually, which is extremely time-consuming, especially if you want to harmonize internationally, or we buy a tool and set up a reasonable process…with the business divisions. This was definitely initiated by BEPS. I personally see this as a great advantage." (INT 1)

Some MNEs were even planning on making the use of the tool mandatory for local subsidiaries, which they believed would enhance consistency in the documentation outputs and efficiency in producing and updating the annual documentation.

"We have the manpower so that we can actually do more, and we did a project last year in which we created templates for the most important transactions, for documentation templates that we will use for the German companies, which we will also give to the foreign companies, and now, in principle, we have also established a process, which is already in place. It has not yet been approved by the Board of Management but by the decisive, let's say, middle management divisions and [company] CFO, and so on. They have already agreed. What counts is that the local companies must basically work with our templates and produce annual documentation…In this respect, we work with templates, but we are only in the process of getting rid of this really seriously." (INT 11)

As stated, one of the benefits of increased consistency in local files is the ability to present standardized local documentation across tax jurisdictions in which the functional profile of the legal entities is similar. Given that global tax authorities are increasingly exchanging information to assess consistency in MNEs' ITP practices throughout their value chains, having a consistent documentation approach was found to be an important driver when considering or using software tools for ITP documentation.

"Ideally, our templates will of course improve from year to year because we intend to review and adapt them, but so does an auditor in Spain who perhaps also talks to an Italian auditor who says that they can, for example, lay the transfer price documentation of both units next to each other and say, 'Well, exactly the same thing here.' So, that is our vision, I would say." (INT 13)

"We are currently converting them (documentation files) based on BEPS. Up until now, it was rather a loose-leaf collection. Now we are really building up a consistent documentation, which is then used uniformly everywhere. Currently, we are still doing it in [Microsoft] Word, but we are in the process of introducing a tool that will take the work off our shoulders and keep the whole thing consistent. At the moment, whenever we submit a local file somewhere, we have to manually check that it is consistent with other local files. This is, of course, incredibly time consuming." (INT 1)

These tools typically offer pre-built text narratives for the documentation files that can be used when local subsidiary controllers and the central tax department collaborate to develop documentation files. Moreover, some tools also have integrated workflow modules that facilitate the central tax department's workflow management for developing ITP documentation. For example, one feature of these tools is the ability to manage the timing of specific actions and communication between in-house tax personnel and business professionals to ensure that the necessary actions are taken timeously, e.g., automated emails to subsidiary finance managers and controllers at certain dates during the documentation cycle to ensure that compliance deadlines are met. Tax departments generally recognize that local controllers and finance managers are not only involved in tax compliance but many other tasks that need to be prioritized. Therefore, such documentation tools were perceived by several interviewees to be valuable in facilitating an organized workflow schedule that is integrated with busy subsidiary-level work schedules.

"What we've actually done is to buy a system that ensures, for example, automatic emails that request people to actively do something…This is something that will be big in the next couple of years because this is how to push the [documentation] processes to life. I am positive but it wońt be easy…To be honest, we are not asking to do more, we are only asking to do it more structured. Sometimes it is difficult for some people when they work in the year-end closing period or the budget phases. If they work 60 hours a week, it is difficult to squeeze in transfer pricing compliance, but we are focusing on the periods in which they are not so busy and making sure that they do it consistently." (INT 9)

Other interviewees explored these documentation tools in demo formats but seemed to be discouraged because they believed that a significant number of resources are needed to customize and maintain these tools.

"We once looked at an online documentation tool that automatically generates documentation from the text modules…I think that it is also extremely time consuming to maintain." (INT 22)

An additional issue that caused some interviewees to be less interested in these tools was that they believed that they would not be able to adequately adapt to the local documentation requirements, which means that a local consultant would still need to be engaged to ensure that the final documentation file was compliant with local rules.

"We still had a local consultant who created transfer pricing documentation in addition to our tool because the documentation, which was tool-based, did not meet local requirements." (INT 13)

Of note, some individual tax authorities are implementing short deadlines for MNEs to respond to information requests during tax audits. In some cases, this results in the manual ITP approach not being viable. One interviewee pointed to South Korea as a jurisdiction in which, in some cases, MNEs' feedback to tax authorities during tax audits needs to be provided within a matter of days. This adds tremendous pressure to the manual data processing approach and incentivizes MNEs to automate the ITP documentation process.

"Korea, for example, has an extremely short review period…You have two or three days between the request from the auditor and the presentation of the documents, if you are lucky." (INT 1)

In general, we observed a more positive and optimistic attitude towards the automation of the documentation tasks relative to the price setting process. The majority of the MNEs studied were either strongly considering automating the creation of the documentation files, including the process workflows, or were actively involved in projects to implement various automation tools to facilitate this process. Of note, while the BEPS project has significantly increased attention from top management on the activities of the tax department, the allocation of resources to deal with the BEPS documentation requirements generally do not seem to have increased accordingly. In this context, it appears that for some MNEs, the decision on whether to convert from a manual documentation approach to a more automated documentation process is driven by more than just the fundamental appeal of automation. Specifically, there seems to be an urgent need for automation to ensure that the scarce tax department personnel can process the increasing number of tasks required of them.

Table 2
Summary of findings.

	Setting transfer prices	**Documenting transfer prices**
General Observations	• The use of information technology for automating price setting and monitoring margins is limited. • Some MNEs are skeptical that a full ITP automation is realistic. • A few MNEs have invested in price setting tools for margin monitoring (ex-post) and transfer price simulation analysis (ex-ante). • Many MNEs engage in manual processing of accounting data and depends on Excel applications. • Tax departments have been subject to increased organizational awareness, particularly from top management teams.	• ITP documentation processes are more automated than ITP price setting • New OECD documentation guidelines create a need for more consistent and standardized documentation file formats. Consequently, MNEs are producing local file templates to enhance standardization across tax jurisdictions. • Several MNEs use ITP documentation tools that include pre-narrated templates, which can be customized to specific business functional circumstances. • ITP documentation tools are mainly used to support workflow management. In contrast, the use of software to process ERP data and the automated importing of accounting information into documentation files are limited.
System-level observations	• ERP system fragmentation significantly reduces the degree of ITP automation. • Standardized group-wide ITP policies and methods, which are perceived to be a prerequisite for automation, are not in place within all MNEs. • Only a few MNEs are in the initial phase of costly long-term ITP–ERP system harmonization and automation projects. Another small group of MNEs are using external price-setting tools that are not integrated with the ERP system. • The implementation process is complex and time-consuming, adding to the currently limited ITP automation that was observed. • The tools are generally perceived as being valuable. However, an effective implementation requires a significant adjustment to the ERP environment and an allocation of additional resources to the tax function. • ERP system designs are based on internal control and management reporting needs. In addition, the system design and functionality are handled by the management accountants or the IT department - not the tax department. • Tax departments play a subordinate role to the more powerful management accountants and in-house IT experts in charge of ERP system design, which limits their ability to automate ITP due to conflicting data segmentation for managerial reporting versus tax reporting.	• ERP system fragmentation reduces the basis for the automation of documentation files. • The format of management reporting reduces the ability to automate ITP documentation. Specifically, management reporting is based on divisional or business unit-level reporting, while ITP tax compliance is based on legal entity (company)-level reporting. • ITP documentation tools generally have strong appeal among MNEs and in many cases, are either being considered or in the process of being implemented. However, in some cases, these tools are considered to be too time consuming and expensive, e.g., if local tax rules necessitate extensive customization of the tools' text templates.
Individual-level observations	• Tax professionals are generally aware of the need for increasing information technology capabilities to enhance the integration of tax compliance in ERP systems. • Some tax professionals are perceived as having limited information technology skills and, in some cases, portrayed as being uncomfortable working with complex applications. • The importance of technology for ITP is generally recognized, and some MNEs are actively employing technically skilled human resources.	• Tax professionals at MNEs generally have a vision of automating the documentation process. • The motivation for automation appears to be partly driven by tax departments' limited human resources for dealing with the increasing documentation requirements as a result of BEPS.

"We have to make sure that we have the right resources and that we have the right system because the compliance burden will increase, and the number of employees will not increase, at least in the tax department." (INT 9)

Table 2 shows a summary of the findings of the empirical analysis discussed above.

5. Discussion

In this paper, we studied MNEs' use of information technology for managing ITP tax risks, including the degree to which they have automated their ITP processes to ensure compliance with the arm's length principle. While our respondents generally viewed ITP automation as an appealing idea, we observed that the ERP system configurations in MNEs create numerous barriers to ITP automation at both the system and individual level. At the system level, the main barriers observed were fragmentation in ERP systems and differences in the accounting information needs for management reporting and tax reporting. Our findings confirmed observations within the contingency school of accounting research that MNEs' tax departments are subject to increased organizational attention (Cools et al., 2008; Cools and Slagmulder, 2009; Plesner Rossing, 2013). However, we discovered that tax departments still play a subordinate role to more powerful in-house accountants and information technology experts in charge of ERP system design. Specifically, ERP system designs were found to be based on the need for internal control and management reporting, while tax compliance objectives still received relatively limited priority in this. Ultimately, the prioritization of management reporting over tax compliance in ERP system design caused tax departments to gravitate towards traditional Excel solutions to handle ITP. When considering the issue of fragmented ERP structures, our results imply that the top management of MNEs should consider ITP as an additional important factor when evaluating the pros and cons of investment in organizational ERP system alignment. Specifically, when conducting a cost–benefit analysis of an ERP system, its ability to reduce ITP tax risks should be considered as an additional benefit.

In contrast to the limited automation observed in transfer price setting, the degree of automation using ITP tools seemed to be more widespread in the documentation process. However, automation using ITP tools is mainly tied to workflow management, and not the automated importing of accounting data into the local documentation files. Our findings showed that tax departments in MNEs use a highly manual documentation process based on Excel models developed in-house, which ultimately reduces data integrity and consistency in ITP outputs. An important implication of this is that the tax department has a reduced ability to accurately demonstrate to tax authorities the process by which accounting data were imported to documentation files. Even though tax authorities are enhancing their digitization of the tax risk assessment and audit processes (EY, 2017), we found that MNEs continue to use manual and potentially error-prone processes. We believe that as the tax authorities continue this digitization, it will drive the expectation that MNEs should also improve their digitization and automation for ITP. Importantly, we believe that the tax authorities will impose stricter requirements on MNEs being able to effectively produce a trail from the accounting data extracted from ERP systems to the intercompany accounting information presented in the documentation. Our research implies that many tax departments in MNEs still need to make significant improvements in their ERP systems before being able to provide such a trail in an efficient and transparent manner. At the most fundamental level, our findings suggest that a standardized ITP policy across the value chain is a prerequisite for effectively automating intercompany transactions within ERP applications.

We also observed several individual-level barriers to ITP automation. Historically, some in-house tax professionals have been reluctant to be involved in the link between ITP and ERP functionality. Based on our research, we observed that tax departments are increasingly recognizing the importance of the complex integration between ITP and ERP systems, particularly in the wake of BEPS. We believe our research implies a significant need for CFOs and senior-level tax directors to initiate more focused efforts to ensure agility in terms of information technology within tax departments. These efforts could entail reassigning in-house experts from the information technology function to the tax department to inject a technological affinity in ITP activities. Another related implication is that the recruitment efforts of tax departments need to focus more explicitly on candidates' technological capabilities, including assessing their direct experience with complex ERP structures and knowledge relating to how ITP policies can be integrated with the organizational ERP infrastructure. We argue that if tax departments are to successfully engage in organizational debates on ERP system design to promote ITP automation, they need to enhance their technological knowledge base.

In the context of Kung et al (2015) and the taxonomy of mimetic, coercive, and normative pressures (Powell and DiMaggio, 1991), we found that the limited degree of ITP automation was primarily driven by the need to reduce manual data processing and improve the workflow management of the documentation process. We conclude that the 2017 revision to the OECD Guidelines, particularly the updated ITP documentation format, poses a significant coercive pressure on MNEs' tax departments and their resources. While ITP automation is not explicitly promoted in the OECD Guidelines, the coercive pressure from domestic OECD-based tax regulations was found to be a key driver for more automated documentation processes.

Our findings contribute to research into the role of management accountants (Byrne and Pierce, 2007; Goretzki et al., 2013; Horton and Wanderley, 2018; Shields, 1997), as we demonstrated their extensive power in MNEs in comparison to in-house tax professionals. While tax departments in MNEs have significantly increased their standing within the organizational hierarchy and are increasingly involved in business strategic decision making, we found that they still play a secondary role in ERP system design decisions. Therefore, our findings echo previous research regarding the notion that management accountants in collaboration with in-house information technology staff play a key role in ensuring that ERP designs provide a basis for organizations to achieve their commercial goals (Grabski et al., 2011; Granlund, 2011; Rikhardsson and Yigitbasioglu, 2018). Based on our findings, we add to this stream of literature by confirming that the powerful position of management accountants continues to exist in MNEs and, importantly, that it exists at the expense of tax strategic objectives. This dynamic was most obvious in relation to the different data segmentation needs for management reporting versus tax compliance.

Finally, we will outline some limitations of this research and provide suggestions for future research. First, our research focused on

the degree of ITP automation in MNEs but did not consider the use of specific ERP software applications or add-on ITP tools in detail. Therefore, future research could conduct a more detailed study of the degree to which specific ERP applications allow for more efficient ITP automation and their compatibility with specific add-on software solutions. This includes exploring the trade-offs between ITP automation via ITP modules integrated in ERP software packages and ITP automation based on add-on ITP software solutions. Second, future work could study whether certain types of value chains, business models, and other organizational attributes fit ITP automation objectives better than others. Third, this research focuses primarily on MNEs based in Germany and tax department professionals employed at organizational headquarters. Future research could explore MNEs based outside of Europe and interact more directly with senior controllers, information technology specialists, and local business unit finance managers to determine how they perceive the opportunities for ITP automation. Finally, it would be interesting to understand the perspectives of external tax consultants and ITP software developers on ITP automation in MNEs. Specifically, the literature could benefit from insights into how these external stakeholders—given their experience and expertise with ITP automation—perceive the ability to reduce the barriers to automation observed in our research.

Appendix A

Interview guideline

I **General information about the interviewee**
1. What is your current job title?
2. To whom do you report in your professional capacity?
   a. Management accounting
   b. Tax
   c. Financial accounting
   d. CFO
   e. Other
3. Where is your department located in the organizational structure? Which decisions are your department authorized to make?
4. How many years of work experience do you have?
   a. 1–3 years
   b. 3–5 years
   c. 5–10 years
   d. >10 years
5. For how many years have you been involved in transfer pricing as part of your professional duties?
   a. 1–3 years
   b. 3–5 years
   c. 5–10 years
   d. >10 years
6. Which of the following transfer pricing areas are you involved in?
   a. Strategy
   b. Price setting
   c. Compliance
7. Do you collaborate with other departments to set transfer prices? If yes, with which departments do you collaborate?

II **General information about the company**
(obtained from annual reports and website research)

III **General questions about transfer pricing transactions**
1. Which intercompany cross-border transactions exist in your company?
2. What is the current cross-border intercompany volume in your company?
   a. Between 200 million and 500 million euros
   b. Between 500 million and 1 billion euros
   c. Between 1 billion and 5 billion euros
   d. Between 5 billion and 10 billion euros
   e. More than 10 billion euros
3. Does your company have a guideline for transfer pricing? What does the guideline entail?
4. How is the guideline implemented?
   a. Transfer prices for business purposes (management accounts)
   b. Transfer prices for tax purposes
   c. Uniform transfer prices for business and tax purposes

IV **General questions about the transfer pricing system's design and process**
1. How are responsibilities for transfer pricing tax compliance allocated across your organization?
   a. Geographically
   b. Transaction-based

   c. Departmental/individual (transfer pricing manager)
2. Who is responsible for developing and implementing the following areas in the process of transfer pricing? Which other functions are involved, and which interfaces exist for (daily) cooperation?
   a. Development of a holistic transfer pricing strategy
   b. Transfer pricing calculation and setting
   c. Creation of a transfer pricing policy
   d. Design of operational processes and purchase/development of IT transfer pricing solutions
   e. Systematic transfer pricing settlement and margin monitoring
   f. Creation of the transfer pricing documentation
   g. Defense and dispute resolution in tax audits
3. How do you evaluate the cooperation between the tax function and other operational functions involved in the process of transfer pricing?
   a. Integrated
   b. Transparent
   c. Improvable
   d. No cooperation
   How could you improve this cooperation?
4. How do you ensure adherence with the arm's length principle for transfer prices when no benchmark studies are available?
5. How do you use information technology (IT) in the process of transfer pricing in your company? Do you use special software for transfer pricing?
   a. ERP
   b. Excel
   c. Other software
6. How do you evaluate the future role of IT in the process of transfer pricing? Which requirements and characteristics of an IT system would be necessary for a holistic and integrated implementation?
7. What are the responsibilities of employees within the transfer pricing function?
8. What training opportunities are provided to them?
9. Do you participate in networking events to interact with other specialists about transfer pricing?

V **General questions about the transfer pricing objectives and the influence of internal and external factors**

1. What are the current objectives of your transfer pricing system, and how are these objectives prioritized?
2. What influence do external factors have on the transfer pricing system? How are these external factors taken into account in the internal design of the transfer pricing system?
   a. Increased focus on tax compliance, especially because of the OECD BEPS project
   b. Different tax systems and practices of tax authorities
   c. Digitalization, customers, suppliers, external accounting rules, market volatility, tariffs, etc.
3. For which internal transactions and in which specific countries did problems occur in previous audits? What were the reasons for these complications?
4. Has the transfer pricing system been changed due to these complications? If so, what measures have been taken?
5. To what extent have the following internal factors impacted past tax audits and the adaption processes in the transfer pricing system?
   a. Historical facts of the organizational structure
   b. Overall corporate strategy
   c. Taxation strategy
   d. Cultural differences between the employees involved
   e. Different skills of the employees involved
   f. Impairment of divisional goals or corporate goals

VI **General questions about planning and evaluating a transfer pricing system**

1. Do you use planning, control, and evaluation tools in the process of transfer pricing?
2. How do you assess your current transfer pricing system vis-à-vis the four objectives below?
   a. Functionality (fulfillment of legal requirements)
   b. Economics (efficiency of resource allocation, evaluation of performance, and cost savings through tax planning)
   c. Organization (optimal balance between differentiation and integration)
   d. Strategy (promotion or influence of corporate strategy through the transfer pricing process)
3. Does your company use key measures to assess transfer pricing managers' performance?
   a. Group tax rate
   b. Other key figures
   c. No evaluation
4. Does your company use performance measures to incentivize tax compliance?
5. How could you use key measures to strike a satisfactory balance between optimal tax planning and compliance? Which types of key measures would be required?

VII **Challenges and opportunities in the current transfer pricing processes**

1. What should be improved in the phases of planning, controlling, and evaluating the transfer pricing process in your company? How could this improvement be achieved?
   a. Planning
   b. Controlling
   c. Performance evaluation

   What are perceived as the future challenges of planning, controlling, and evaluating your transfer pricing process?
2. What would be required to achieve an integrated and holistic transfer pricing process? Could you give three characteristics of an optimal transfer pricing system and justify your choices?
3. Would you say that the BEPS project brings opportunities for your business? If so, how could you leverage those opportunities?

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